$$e^+ + e^- \longrightarrow 2\gamma \qquad E_\gamma \approx mc^2$$

$s = 1/2$ $\quad s = 1/2$ $\quad S = 0$ $\quad 25\%$

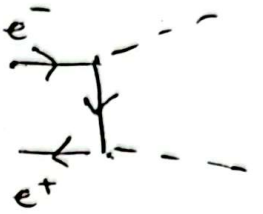
$(S_z = 0)$

$\sim 10^3$ times more likely.

$$e^+ + e^- \longrightarrow 3\gamma$$

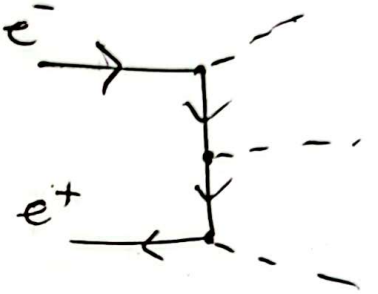
$S = 1$

$(S_z = -1, 0, 1)$



$$\boxed{e^+ + M \longrightarrow M^+ + 2\gamma} \quad (*)$$

direct (in flight)



Dirac: $\sigma_{ann} = \pi r_0^2 \dfrac{c}{v}$

c - speed of light

v - collision velocity

r_0 : classical electron radius

$$\frac{e^2}{r_0} = mc^2$$

(CGS) $r_0 = \dfrac{e^2}{mc^2}$

$r_0 \sim 10^{-4}$

$$\sigma_{ann} \sim \frac{\pi}{c^3} \cdot \frac{1}{v}$$

$$= 10^{-6} \frac{1}{v}$$

$v \sim ?$ $\quad v = 1.$

$\dfrac{v^2}{2}$ $\quad E = \frac{1}{2}$ a.u.

$= 13.6$ eV.

1 a.u. $[E] = 27.2$ eV

Atomic units:

$e = 1$, $m = 1$

Fine-structure constant: $\hbar = 1$

$$\frac{e^2}{\hbar c} \approx \frac{1}{137}$$

$e^2 = E \cdot L = E \cdot T \cdot \dfrac{L}{T}$

$c \approx 137$ a.u.

~~Et~~ For (*):

$$\sigma_{ann} = \pi r_0^2 \frac{c}{v} Z_{eff}$$

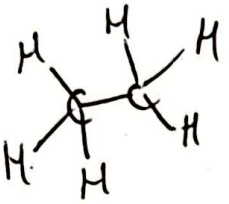
$Z_{eff} \sim Z$ (actual no. of e^-) ???

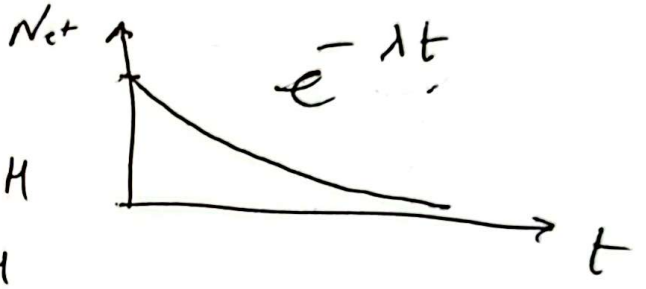
N_2	$Z_{eff} \sim 30$.

Martin Deutsch
1950's

~1995 (Surko)

CH_4	154
C_2H_6	660
C_3H_8	3500
C_4H_{10}	10 000
C_5H_{12}	36 000
C_6H_{14}	120 000
⋮	
$C_{12}H_{26}$	$\sim 10^6$
⋮	
C_nH_{2n+2}	





$$\lambda = \sigma_{ann} \cdot n \cdot v$$

L^2 $\quad \frac{1}{L^3}$ $\quad \frac{L}{T}$

number density

α_d - dipole polarisability

$$U \simeq -\frac{\alpha_d e^2}{2r^4}$$

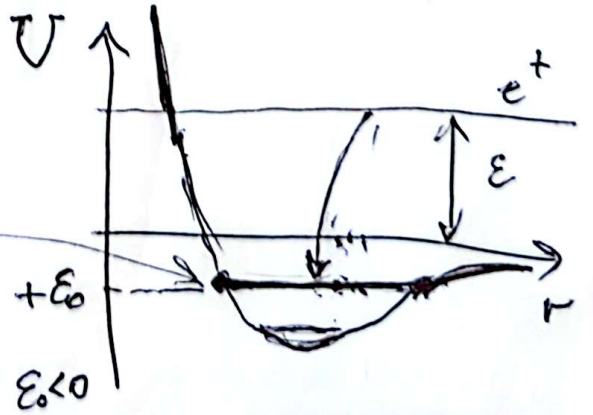
~ 2000

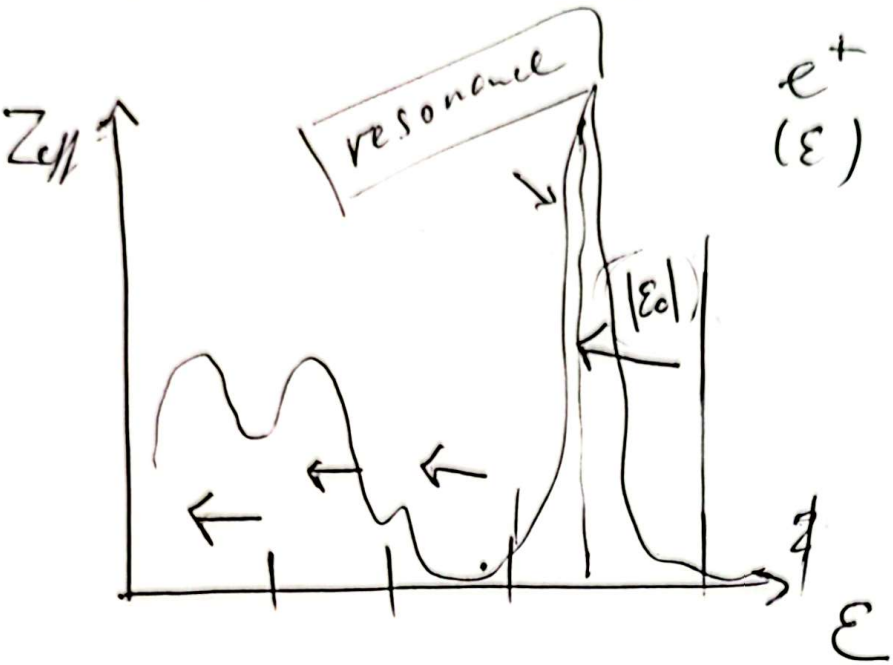
resonant

$$e^+ + M \rightleftarrows e^+M \rightarrow M^+ + 2\gamma$$

ε

$$\varepsilon + |\varepsilon_0| = \hbar\omega_\nu$$





~2003

$$\varepsilon = \hbar\omega_r - \underbrace{|\varepsilon_0|}_{\text{binding energy}}$$

Vibrational Feshbach resonance

$$M_C \approx 12 M_H \approx 12 M_p = \underbrace{12 \times 1836}_{10^4} m_e$$

$10^4 m_e$.

What is the mechanism of e^+ exciting molecular vibrations? (how?)

Rev. Mod. Phys 2010.

Web of Science

References (look back)
Citations (look forward)